



6. Resistance and Capacitance

Only questions about the results of experiments have a real significance and it is only such questions that theoretical physics has to consider.

– P. A. M. Dirac, *The Principles of Quantum Mechanics*, 4th ed. (1958), §2.

6.1 Steady Current

When an external electric field is applied, a force is exerted on the conduction electrons inside a conductor, leading to a macroscopic motion of electric charges, called an electric current. Since both electric charge q and time t are scalars, current I is a scalar as well. The current I is defined as the charge that passes through a hypothetical plane per unit time, or:

$$I = \frac{dq}{dt}$$

The charge passing through a specific plane during time period t_1 to t_2 is therefore:

$$q = \int_{t_1}^{t_2} I dt$$

We shall focus on a special case of electric current, namely **steady current**, meaning the macroscopic property of the current does not vary in time. Thus, for a steady current, the current through every plane that completely cuts the conductor is the same at all times.

Although the current is a scalar, we often assign an arrow for it in a circuit diagram. The arrow is simply used for specifying how the charge is moving. It is conventional to draw the arrow in the direction in which positive charge carriers would move to form this current, although the actual charge carrier could be negative and move in the opposite direction.

If we have a non-uniform conductor or an irregularly shaped conductor, although the current through different cross-sections of the conductor stays the same because of charge conservation, its spatial distribution may be very different.

Assuming we take a small vector area element $\delta\mathbf{S}$ at $\mathbf{r}$, there is in general a fraction of the current δI passing through it. We define the **current density** $\mathbf{j}$ at $\mathbf{r}$ as:

$$\mathbf{j} = \max_{\hat{n}} \left\{ \lim_{|\delta\mathbf{S}| \rightarrow 0} \frac{\delta I}{|\delta\mathbf{S}|} \right\} \hat{n}$$

where the unit vector $\hat{n}$ is taken along the $\delta\mathbf{S}$ that gives the maximum value.

Therefore, the current I through any surface S can be calculated as:

$$I = \int_S \mathbf{j} \cdot d\mathbf{S}$$

It is helpful to compare the relation between the current and the current density with the one between electric flux Φ and the electric field $\mathbf{E}$.

6.2 Resistance and Ohm's Law

Macroscopically, when a potential difference V is applied to a conductor, a current I forms. We define the resistance R of a conductor as:

$$R = V/I$$

Conductors of the same shape but different materials can have different resistance. Therefore, we also define the **resistivity** ρ_e and the **conductivity** σ of a material. The two are related by $\rho_e = \sigma^{-1}$, and they serve as an indication of how good (or bad) a conductor is: $\mathbf{j} = \sigma\mathbf{E}$.

Microscopically, when a potential difference is applied to a conductor, an electric field builds up inside. This causes the charge carriers in the conductor to start picking up additional velocity in addition to their thermal random motion. This additional velocity is called the **drift velocity** $\mathbf{v}_d$ (Figure 6.1). For electrons, whose charge is $-e$, it is opposite to the field; positive carriers drift along it.

Question: Can you estimate the speed of the electronic thermal random motion at room temperature (300K)?

Equating the thermal energy to the kinetic energy of a free electron gives $v_{th} = \sqrt{3k_B T/m} \approx 1.17 \times 10^5$ m/s at 300 K — nine orders of magnitude larger than the drift velocity of a typical current. This is the speed of a classical gas; the conduction electrons of a metal move more than ten times faster, for a reason that needs quantum mechanics.

For simplicity, we assume a cylindrical conductor with cross-sectional area S and length L . The conductor is made of a uniform material with a number density of charge carriers (for example, electrons) n . A potential difference V applied across its ends generates a current I through this conductor. Imagine a cross-section of this conductor: $N = nSv_d\Delta t$ carriers pass through it within

time Δt , where $v_d = |\mathbf{v}_d|$. Thus, the magnitude of the current is $I = Ne/\Delta t = neSv_d$. Since the conductor is uniform, we immediately have:

$$\mathbf{j} = -ne\mathbf{v}_d$$

Now, for this simple system, we take a closer look at how $\mathbf{v}_d$ is related to the electric field $\mathbf{E}$. One thing we learn from thermodynamics is that particles are always in random motion and constantly collide with each other. In this classical picture the collisions that matter are those of the electrons with the ions of the metal. A collision between two electrons conserves their total momentum, and so cannot by itself change the current. The collisions are so frequent that the effect of $\mathbf{E}$ is not to dramatically change its speed. Instead, the electrons will only pick up some small deviation due to the field during two consecutive collisions.

Since there are a huge number of collisions in all kinds of directions, we can assume that on average, each electron will “lose all its memory” of its previous velocities, starting fresh and moving off in a random direction. Let the average time between two consecutive collisions be the **mean free time** τ , and suppose that an electron has the same probability per unit time of colliding, however long since its last collision. Then at any moment the average time since its last collision is τ , and the velocity the electron has gained from the field, averaged over all electrons, is the drift velocity:

$$\mathbf{v}_d = -\frac{e\mathbf{E}}{m}\tau$$

Together with the previous expression for $\mathbf{j}$, we shall have:

$$\mathbf{j} = \frac{ne^2\tau}{m}\mathbf{E}$$

Therefore, $\sigma = \frac{ne^2\tau}{m}$ from our derivations above. For normal metals under normal conditions, both n and τ can be considered to be constants, which is Ohm’s law.

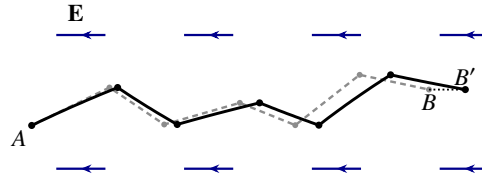


Figure 6.1: The motion of conduction electrons in a metal under an applied electric field $\mathbf{E}$, which points to the left. The dashed gray path is a sequence of free flights with the field switched off; the solid black path is the same sequence with the field on, so that every flight is bent opposite to the field. The dots mark collisions, at which an electron “loses memory” and moves off in a new random direction. The displacement that accumulates along the path is the drift velocity $\mathbf{v}_d$, an average over all electrons.

Using $I = jS$, $V = EL$ and $R = V/I$ together with $j = \sigma E$, the resistance of a uniform conductor of cross-section S and length L follows as:

$$R = \rho_e \frac{L}{S}$$

Question: Suppose, classically, that an electron’s mean free path does not depend on temperature, and that its speed is the thermal speed estimated above. How do τ and the resistivity then depend on T ? Compare with the measured, nearly linear rise of a metal’s resistivity near room temperature.

Classically the mean free time is $\tau = \lambda/v_{\text{th}}$, where λ is the **mean free path** (the average distance an electron travels between two consecutive collisions), which we take to be independent of temperature, and $v_{\text{th}} \propto \sqrt{T}$, so $\tau \propto T^{-1/2}$ and the resistivity $\rho_e = m/(ne^2\tau) \propto \sqrt{T}$: it should rise as the square root of temperature, whereas a real metal's resistivity rises nearly linearly near room temperature. Real metals do not obey this law, which is one of the reasons the classical picture has to be replaced by quantum mechanics.

Finally, charge is conserved in every known process; even the creation or annihilation of a pair of particles adds or removes equal amounts of positive and negative charge. Hence the charge inside a fixed closed surface $\mathcal{S}$ changes only by the current through it, which in a mathematical expression:

$$\frac{d}{dt} \int_V \rho \, dV + \oint_{\mathcal{S}} \mathbf{j} \cdot d\mathbf{S} = 0$$

where the volume integration takes place within $\mathcal{S}$. By the divergence theorem of Section 4.2 the surface integral equals $\int_V \nabla \cdot \mathbf{j} \, dV$ over the same volume, and since the volume is arbitrary, we can see immediately:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{j} = 0$$

6.3 Capacitors and Capacitance

Imagine we now have two conductors. If they are separated and isolated from each other, and one of them is charged with $+q$ and the other with $-q$, there must be a potential difference V between them. If a conducting wire is connected to both conductors, a current will immediately flow through the wire, bringing both of them to charge neutrality. We define a **capacitor** as two isolated conductors that can hold a potential difference V when charged with opposite charge $+q$ and $-q$, and its **capacitance** C as:

$$q = CV$$

We shall see that, like R , C is in most cases a constant depending only on the geometry (shape and relative position) of the conductors and the material in between. For conductors of fixed shapes and positions with vacuum between them, doubling the charges on both conductors doubles the field, which by superposition is again an equilibrium and, by the uniqueness of Chapter 5, the only one; so V doubles. Hence q/V does not depend on q , only on the shapes and positions of the conductors. With a linear dielectric between them the same holds, by a uniqueness theorem for that case that we do not prove.

Example 6.1: Capacitance per area of an infinitely large parallel-plate capacitor

Let's assume this capacitor consists of two infinitely large parallel metallic plates at $x = -d/2$ and $x = d/2$, the left one charged to a surface charge density σ_s and the right one to $-\sigma_s$ (Figure 6.2). For plates this large, the field between them is σ_s/ϵ_0 , uniform and perpendicular to the plates, and it is zero outside them; therefore the potential drop between the two plates is:

$$V = Ed = \frac{\sigma_s}{\epsilon_0} d$$

The capacitance per unit area, C/S , is therefore, using $q = \sigma_s S$:

$$\frac{C}{S} = \frac{q}{VS} = \frac{\sigma_s}{V} = \frac{\sigma_s}{\frac{\sigma_s}{\epsilon_0} d} = \frac{\epsilon_0}{d}$$

This is exactly the formula for a parallel plate capacitor that we are familiar with:

$$C = \frac{\epsilon_0 S}{d}$$

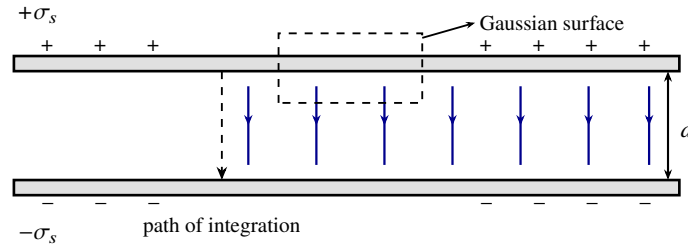


Figure 6.2: A parallel-plate capacitor in cross-section: two plates of area S , running into the page, separated by a distance d and carrying the surface charge densities $+\sigma_s$ and $-\sigma_s$. The dashed rectangle is the Gaussian surface whose flux gives the field between the plates, and the dashed vertical line is the path along which that field is integrated to give the potential difference.

Therefore, in order to calculate the capacitance of a specific capacitor, we have to 1) assume equal and opposite charges on the two conductors, 2) find the electric field they produce between them, 3) integrate the field along a path between the conductors to obtain the potential difference, and 4) calculate C using $C = q/V$.

Example 6.2: Capacitance per length of an infinitely long cylindrical capacitor

We shall assume that this capacitor consists of two infinitely long coaxial cylinders of radii a and $b > a$. If the inner cylinder is charged to a linear charge density λ and the outer one to $-\lambda$, the electric field between the cylinders is radial (Figure 6.3) and can be easily obtained from Gauss' law, Eq. (2.1):

$$E = \frac{1}{2\pi\epsilon_0} \frac{\lambda}{r}$$

This immediately yields the potential difference between the two cylinders:

$$V = \frac{1}{2\pi\epsilon_0} \lambda \ln\left(\frac{b}{a}\right)$$

Thus the capacitance per length:

$$\frac{C}{L} = \frac{2\pi\epsilon_0}{\ln\left(\frac{b}{a}\right)}$$

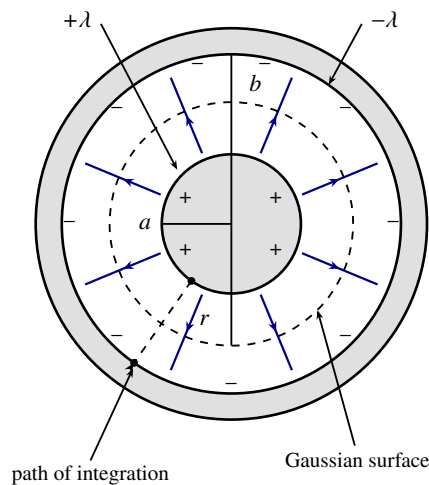


Figure 6.3: A cylindrical capacitor in cross-section: two coaxial cylinders of radii a and b , the inner one carrying the linear charge density $+\lambda$ and the outer one $-\lambda$, with a radial field between them. The dashed circle of radius r is the Gaussian surface of Example 6.2, and the dashed radial line is the path along which the field is integrated to give the potential difference.

Example 6.3: Capacitance between a single conducting sphere and infinity

This set-up may initially sound quite strange, but you can imagine the second plate to be a concentric spherical shell, whose radius is taken to infinity. Assuming that the sphere has radius a and is charged to q , the electric field outside the sphere is simply (Figure 6.4):

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

Thus, the potential difference between the sphere and infinity, taking $V(\infty) = 0$, is

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{a}$$

With this, we immediately obtain the capacitance to be:

$$C = \frac{q}{V} = 4\pi\epsilon_0 a$$

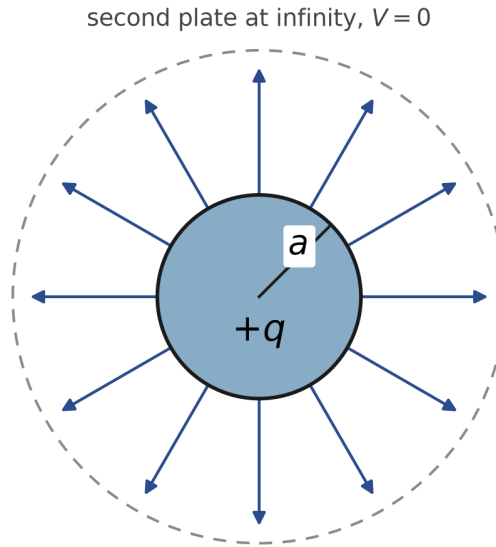


Figure 6.4: A single conducting sphere of radius a charged with q . The second plate is taken to be a concentric spherical shell at infinity, giving $C = 4\pi\epsilon_0 a$.

6.4 Dielectrics*

Previously, we have considered electric field inside a metal. We see that under electrostatic equilibrium condition, $\mathbf{E} = 0$ should be satisfied everywhere inside the metal. Such an effect is called “screening”. In semiconductors or insulators, the “screening” is not sufficiently strong and thus the electric field can still penetrate them. In principle, we can still use Gauss’ law or Poisson’s equation, Eq. (4.3), to solve for the electric field inside an insulating material. However, if we apply the original Gauss’ law, we must take into account any microscopic charge distribution (that is varying at the scale of $1 \text{ \AA}$). The resulting microscopic electric field $\mathbf{E}_m$ also varies abruptly at small scale. However, the information about the *microscopic* field is not helpful if we are interested in the *macroscopic* properties. The latter is usually related to an average of a microscopic quantity, namely $\langle \mathbf{E}_m \rangle$. In practice, this quantity is much more useful, and is referred to as the electric field

inside the material $\mathbf{E} = \langle \mathbf{E}_m \rangle$. Experiments have shown that the electric flux of $\langle \mathbf{E}_m \rangle$ through a closed surface is still proportional to the **free charge** inside the closed surface, but with a different coefficient than ϵ_0 , namely:

$$\oint \mathbf{E} \cdot d\mathbf{S} = \frac{1}{\epsilon} q_{\text{free}}$$

This form holds when $\mathbf{D} = \epsilon \mathbf{E}$ with one and the same ϵ at every point of the surface, as we shall see below. Keep in mind that the field $\mathbf{E}$ here is no longer the microscopic field, but an average instead. The coefficient $\epsilon = \epsilon_r \epsilon_0$ is the **permittivity** of the material, and the dimensionless ratio $\epsilon_r = \epsilon / \epsilon_0$ is its **dielectric constant** (equivalently, its **relative permittivity**). Thus, $\epsilon_r = 1$ for vacuum. To further simplify the calculations, we introduce:

$$\mathbf{D} = \epsilon \mathbf{E}$$

$\mathbf{D}$ is called the **electric displacement vector**; its meaning in terms of the dipoles of the material is explained below. With it, Gauss' law in different dielectric materials, as well as in vacuum, can be written in one and the same form, of which the equation for $\mathbf{E}$ above is the special case:

$$\oint \mathbf{D} \cdot d\mathbf{S} = q_{\text{free}}$$

and

$$\nabla \cdot \mathbf{D} = \rho$$

Note that both q_{free} and ρ refer to free charge, not to bound charge. The bound charge is the charge of the polarized material; one can show that its volume density is $-\nabla \cdot \mathbf{P}$ and its surface density $\mathbf{P} \cdot \hat{n}$, with $\hat{n}$ the outward normal, so that in a uniformly polarized slab only the surface charge is left.

Therefore, in order to solve for the averaged electric field inside a dielectric material, we first find $\mathbf{D}$ from the free charge: when the symmetry of the free charges and of the dielectric fixes the direction of $\mathbf{D}$ and its magnitude over a Gaussian surface, $\oint \mathbf{D} \cdot d\mathbf{S} = q_{\text{free}}$ gives $\mathbf{D}$, and then $\mathbf{E} = \mathbf{D} / \epsilon$.

Here $\mathbf{E}$ is the averaged field, to which the dipoles of the material contribute as well, so that $\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}$. Microscopically, we can imagine that the originally charge-neutral molecules will be affected by the electric field, so that their positive and negative charge centers become different, and form an electric dipole. In a metal in electrostatic equilibrium the free electrons move until the field inside vanishes, so this description is used for insulating materials, or **dielectrics**. We call $\mathbf{P} = \mathbf{D} - \epsilon_0 \mathbf{E}$ the **polarization**; one can show that it equals the electric dipole moment per unit volume of the polarized material.

Example 6.4: A parallel-plate capacitor filled with a dielectric

Take the capacitor of Example 6.1 and fill the gap with a dielectric of permittivity $\epsilon = \epsilon_r \epsilon_0$. The free charge $\pm \sigma_s$ on the plates is unchanged, so $D = \sigma_s$ still, and $E = D / \epsilon = \sigma_s / (\epsilon_r \epsilon_0)$. Hence $V = Ed = \sigma_s d / (\epsilon_r \epsilon_0)$ and

$$\frac{C}{S} = \frac{\epsilon_r \epsilon_0}{d} = \epsilon_r \left. \frac{C}{S} \right|_{\text{vac}}$$

The capacitance is multiplied by ϵ_r .

6.5 Exercises

1. A copper cylinder has radius a and length l . It is surrounded by a coaxial copper tube of equal length. The inner radius of the copper tube is b ($l \gg b$, so edge effects are neglected). The space between the cylinder and the tube is filled with a uniform material of conductivity σ ; the copper conducts far better than this material, so treat both copper pieces as perfect conductors. Find the resistance R between the cylinder and the tube.
2. A copper cylinder has a cross-sectional area of 1000 mm^2 and length 2.0 m . The potential difference between the two ends of the cylinder is 100 mV . The conductivity of copper is $\sigma = 5.7 \times 10^7 \text{ S/m}$, and the number density of charge carriers is $n = 8.49 \times 10^{28} \text{ m}^{-3}$ (equivalently, the magnitude of the carriers' charge density is $ne = 1.36 \times 10^{10} \text{ C/m}^3$). Compute the following quantities:
 - (a) The resistivity ρ_e ;
 - (b) The current I ;
 - (c) The current density j ;
 - (d) The electric field E inside the copper cylinder;
 - (e) The drift velocity v_d of the electrons.
3. Consider three thin coaxial metallic tubes of the same length l , and radii a, b, c ($a < b < c$). The space between the conductors is vacuum. The inner and outer tubes are connected via a conducting wire. For simplicity, we ignore the edge effects (assuming $l \gg c$). Calculate the capacitance C between the middle tube and the other two.
4. Two parallel long wires are separated by a distance d measured between their axes, and each wire has a radius a ($d \gg a$). The space between them is vacuum. Calculate the capacitance per unit length. (*Hint:* replace the two wires by two line charges $\pm\lambda$ whose equipotential surfaces coincide with the wire surfaces.)
5. Consider a long wire of radius a parallel to an infinite metallic plate. The plate is grounded ($V = 0$). The distance from the axis of the wire to the plate is h ($h \gg a$), and the space between them is vacuum. Calculate the capacitance per unit length. (*Hint:* use the method of images of Chapter 5: the image of a line charge $+\lambda$ at height h above a grounded plane is a line charge $-\lambda$ at depth h , and the potential of a line charge is $V = -(\lambda/2\pi\epsilon_0) \ln r + \text{const.}$)
6. A uniform material contains two kinds of mobile charge carrier. The positive carriers have charge $+e$, number density n_+ , mass m_+ and mean free time τ_+ ; the negative carriers have charge $-e$, number density n_- , mass m_- and mean free time τ_- . All these quantities are positive. A uniform electric field of magnitude E_0 points along the $+x$ direction in the material. Assume that each kind has the drift velocity given by the result of Sec. 6.2, with its own charge, mass and mean free time, and that the two kinds do not affect each other.
 - (a) Find the drift velocities $\mathbf{v}_+$ and $\mathbf{v}_-$ of the two kinds of carrier.
 - (b) Find the current density $\mathbf{j}$ and the conductivity σ of the material. Under what condition do the two kinds of carrier contribute equally to the current density?

7. Two long rods with the same cross-sectional area S , made of materials with conductivities σ_1 and σ_2 , are joined end to end along the x axis. A steady current $I > 0$ flows in the $+x$ direction, from rod 1 into rod 2. Assume that the current density is uniform over every cross-section of each rod, and take the permittivity of both materials to be ϵ_0 .
- Find the electric field in each rod. Why can the fields differ when the current is the same?
 - Find the free charge Q on the interface between the two rods. For which ordering of σ_1 and σ_2 is Q positive?
8. A metal hemisphere of radius a is buried in level ground with its flat face in the ground's surface. The ground is a uniform material of conductivity σ , much smaller than that of the metal, and the air above it does not conduct. A steady current I flows from the hemisphere into the ground and returns through an electrode very far away; in the ground it flows radially outward from the center of the flat face. Let r be the distance from that center, and take the potential to be zero far away.
- Find the current density $\mathbf{j}$ and the electric field $\mathbf{E}$ in the ground at a distance $r > a$, and the resistance R between the hemisphere and the distant electrode.
 - A person stands on the ground with one foot at distance r from the center and the other at distance $r + s$, on the same radial line. Find the potential difference $V(r) - V(r + s)$ between the two feet. Why is it safer to keep the feet close together near such an electrode?
9. A material of uniform conductivity σ obeys $\mathbf{j} = \sigma\mathbf{E}$; take its permittivity to be ϵ_0 .
- Using the continuity equation of Sec. 6.2 and Gauss' law in differential form (Chapter 4), show that a free charge density $\rho(\mathbf{r}, 0)$ present in the material at $t = 0$ becomes $\rho(\mathbf{r}, t) = \rho(\mathbf{r}, 0)e^{-t/\tau_q}$ at every point, and find the time constant τ_q .
 - A ball of this material, of radius b , is isolated in vacuum and carries no charge. At $t = 0$ a charge Q is placed on a small metal bead of radius a at its center, in contact with the material. Find the current $I(r, t)$ flowing outward through a concentric sphere of radius r , where $a < r < b$, and the charge $Q_s(t)$ on the surface of the ball.
10. A parallel-plate capacitor has plates of area S a distance d apart, carrying charges $+q$ and $-q$. Neglect edge effects.
- A slab of dielectric of dielectric constant (relative permittivity) ϵ_r and thickness $w < d$, with the same area as the plates, is placed between the plates and parallel to them. Find D and E in the empty part of the gap and in the slab, and the capacitance C . Does C depend on where the slab is placed?
 - The slab is removed, and a block of the same dielectric, of thickness d and area $S/2$, is placed so that it touches both plates and fills half of the space between them. The plates remain equipotentials, and in each half the field is uniform and perpendicular to the plates. Find the capacitance.